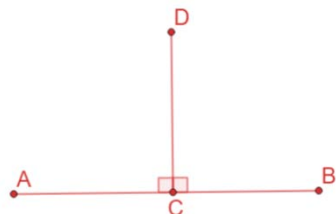


Geometry
Unit 2
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

In questions 1-3, determine if each of the following represents a definition, postulate, or theorem. Then, draw a diagram using the given information for each representation.

1. If two supplementary angles are congruent, then they are right angles.



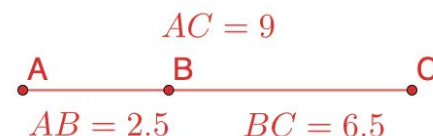
Theorem
(Sketch answers may vary)

2. A midpoint divides a segment into two congruent parts.



Definition
(Sketch answers may vary)

3. If a point, B , is drawn on a line segment AC , then AC is the sum of AB and BC .



Postulate
(Sketch answers may vary)

Complete the table.

	Property	Equality	Congruence
4.	Transitive	If $m\angle ABC = m\angle DEF$ and $m\angle DEF = 45^\circ$, then $m\angle ABC = 45^\circ$	If $\angle ABC \cong \angle DEF$ and $\angle DEF \cong \angle GHI$, then $\angle ABC \cong \angle GHI$
5.	Reflexive	$m\angle JKL = m\angle JKL$	$\angle JKL \cong \angle JKL$
6.	Symmetric	If $m\angle XYZ = 81^\circ$, then <u>$81^\circ = m\angle XYZ$</u>	If $\angle XYZ \cong \angle FGH$, then $\angle FGH \cong \angle XYZ$

7. Write the conclusion for the statement.

If $m\angle JKL + m\angle SMW = m\angle XYZ + m\angle DNF$ and $m\angle XYZ = m\angle SMW$, then...
 $m\angle JKL + m\angle SMW = m\angle SMW + m\angle DNF$,

8. Name the property to justify the statement in question 7.

Substitution Property of Equality

Complete the proof.

	Statement	Reason
9.	$m\angle MNP = m\angle DFG$	Given
10.	$\angle MNP \cong \angle DFG$	Definition of Congruence

Geometry

Unit 2

Lesson 2 Homework

Name **Answer Key**

Date _____

Period _____

In questions 1-5, determine the type of reasoning for each statement.

1. In order to participate in school sports, students must have a minimum GPA of a 2.0. Jim has a GPA of 2.75, therefore he can participate in school sports.

Deductive Reasoning

2. Every Tuesday that Yolanda's family has visited a specific local restaurant, the kids' meals have been free. When Yolanda's family visits that restaurant next Tuesday, the kids' meals will be free.

Inductive Reasoning

3. A new movie director has grossed over 250 million for each of his first three movies. When his fourth movie is released, it will gross over 250 million.

Inductive Reasoning

4. Any given number multiplied by one equals itself. If 5 is multiplied by one, it will equal itself.

Deductive Reasoning

5. The air conditioner is set to keep a particular home at 72°F and runs off the home's electricity. During a power outage, the home will not stay at 72°F.

Deductive Reasoning

For each statement in 6-10, write a counterexample that disproves the statement.

6. Mary enters her home and her clothes are wet. Therefore, it is raining outside.

Answers may vary.

The sprinklers are running outside and Mary walked in their path.

7. If a number is multiplied to another number, it will result in a larger number.

Answers may vary. $5 \times 0 = 0$

8. If a figure has four sides and all sides have the same length, it is a square.

A rhombus has four sides with the same length and is not a square.

9. If a number is prime, then it is an odd number.

2

10. Yvette lives in Jacksonville. Therefore, she lives in Florida.

Answers may vary. Yvette lives in Jacksonville, Alabama.

Geometry

Unit 2

Lesson 3 Homework

Name **Answer Key**

Date _____

Period _____

In the game of “Crazy Eights”, each player is dealt 7 cards. The remainder of the stack is placed face down on the table. The top card from the stack is flipped over and placed face up on the table. The first player must either match the suit (Diamonds, Clubs, Spades, or Hearts), the number (1,2,3,4,5,6,7,8,9,10) or the face card (Jack, Queen, King, Ace).

Using these rules, complete the table to end at a King of Clubs.

Answers may vary. Sample answer shown below.

	Statement	Reason
1.	Ace of Spades	Given
2.	Jack of Spades	Same Suit
3.	Jack of Clubs	Same Face Card
4.	3 of Clubs	Same Suit
5.	3 of Hearts	Same Number

	Statement	Reason
6.	Queen of Hearts	Same Suit
7.	Queen of Diamonds	Same Face Card
8.	9 of Diamonds	Same Suit
9.	9 of Clubs	Same Number
10.	King of Clubs	Same Suit

Geometry

Unit 2

Lesson 4 Homework

Name **Answer Key**_____

Date_____

Period_____

For questions 1-3, use the quote shown.

"If you build it, they will come" – Kevin Costner from the film, *Field of Dreams*

1. Write the converse of this statement.

If they will come, then you build it.

2. Write the inverse of this statement.

If you don't build it, they won't come.

3. Write the contrapositive of this statement.

If they won't come, then you don't build it.

For questions 4-6, use the quote shown.

"When you're curious, you find lots of interesting things to do" – Walt Disney

4. Write the converse of this statement.

If you find lots of interesting things to do, then you're curious.

5. Write the inverse of this statement.

When you're not curious, you don't find lots of interesting things to do.

6. Write the contrapositive of this statement.

If you don't find lots of interesting things to do, then you're not curious.

For questions 7-8, use the quote shown.

"If I had asked people what they wanted, they would have said faster horses". - Henry Ford

7. What is the hypothesis?

I had asked people what they wanted.

8. What is the conclusion?

People would have said faster horses.

Rewrite the following statements as conditional statements.

9. When the going gets tough, the tough gets going.

If the going gets tough, then the tough get going.

10. When you are not first, you are not last.

If you are not first, then you are not last.

Complete each table for the conditional statement.

If an angle is classified as a right angle, then it has a measure of 90° .			
	Relationship to Conditional	Statement	True or False
1.	Converse	If an angle has a measure of 90° , then it is a right angle.	True
2.	Inverse	If an angle is not classified as a right angle, then it does not measure 90° .	True
3.	Contrapositive	If an angle does not measure 90° , then it is not a right angle.	True
Write the conditional using "if and only if".			
4.	An angle is classified as a right angle if and only if it measures 90° .		

If two angles have the same measure, then they are congruent			
	Relationship to Conditional	Statement	True or False
5.	Converse	If two angles are congruent, then the angles have the same measure.	True
6.	Inverse	If two angles do not have the same measure, then they are not congruent.	True

7.	Contrapositive	If two angles are not congruent, then the angles do not have the same measure.	True
Write the conditional using “if and only if”.			
8.	Two angles have the same measure if and only if the angles are congruent.		

9. Write a conditional statement that is biconditional.

Answers may vary. A possible correct answer: If the sum of the interior angles of the polygon is 180° , then the polygon is a triangle.

10. Write a conditional statement that is **not** biconditional.

Answers may vary. A possible correct answer: If it is raining, then there are clouds in the sky.

Geometry
Unit 2
Lesson 6 Homework

Name **Answer Key** _____
Date _____
Period _____

Determine if each statement is true. Rewrite each false statement using the word "some".

	Statement	True or False	Re-write
1.	All even numbers are composite.	False	Some even numbers are composite.
2.	All points that lie in the same plane are coplanar.	True	
3.	All linear pairs are supplementary.	True	

Write each statement as an if-then statement.

	Statement	If-Then
4.	All even numbers are composite.	If the numbers are even, then the numbers are composite.
5.	All points that lie in a plane are coplanar.	If the points lie in a plane, then the points are coplanar.

6.	All linear pairs are supplementary.	If angles are a linear pair, then the angles are supplementary.
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Write the negation of each statement.

	Statement	Negation
7.	$\triangle ABC$ is an equilateral triangle.	$\triangle ABC$ is not equilateral triangle.
8.	$\angle DEF$ and $\angle KLM$ are supplementary angles.	$\angle DEF$ and $\angle KLM$ are not supplementary angles.
9.	$\angle JKL$ and $\angle XYZ$ are vertical angles.	$\angle JKL$ and $\angle XYZ$ are not vertical angles.
10.	$m\angle DEF + m\angle KLM = 125^\circ$	$m\angle DEF + m\angle KLM \neq 125^\circ$

Geometry
Unit 2
Lesson 7 Homework

Name Answer Key
Date _____
Period _____

Complete the proof.

Given: $AB = CD$

Prove: $AC = BD$



	Statement	Reason
1.	$AB = CD$	Given
2.	$BC = BC$	Reflexive Property of Equality
3.	$AB + BC = BC + CD$	Addition Property of Equality
4.	$AB + BC = AC;$ $BC + CD = BD$	Segment Addition Postulate
5.	$AC = BD$	Substitution Property of Equality

Complete the paragraph proof.

Given: $\angle 1$ and $\angle 2$ are complementary and $\angle 2$ and $\angle 3$ are complementary.

Prove: $\angle 1 \cong \angle 3$

It is 6. given that $\angle 1$ and $\angle 2$ are complementary and $\angle 2$ and $\angle 3$ are

complementary. By the definition of 7. complementary angles,

$m\angle 1 + m\angle 2 = 90^\circ$ and $m\angle 2 + m\angle 3 = 90^\circ$. By the 8. transitive property,

$$m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3.$$

Then, $m\angle 1 =$ 9. $m\angle 3$ by the subtraction property. Therefore, $\angle 1 \cong \angle 3$ by the

10. definition of congruence.

Geometry
Unit 2
Lesson 8 Homework

Name _____
Date _____
Period _____

Complete the two-column proof.

Given: $\overline{AB} \cong \overline{CD}$

Prove: $\overline{AC} \cong \overline{BD}$



	Statement	Reason
1.	$\overline{AB} \cong \overline{CD}$	Given
2.	$\overline{BC} \cong \overline{BC}$	Reflexive Property
3.	$AB = CD$ $BC = BC$	Definition of Congruence
4.	$AB + BC = AC$ $CD + BC = BD$	Segment Addition Postulate
5.	$CD + BC = AC$	Substitution Property
6.	$AC = BD$	Substitution Property
7.	$\overline{AC} \cong \overline{BD}$	Definition of Congruence

A partially completed two-column proof is shown. Match each statement on the left with the correct reason to complete the two-column proof by using reasons provided in the bank. Some reasons may be used more than once.

Given: W is the midpoint of $\overline{VX}$, Z is the midpoint of $\overline{RP}$, and $\overline{VX} \cong \overline{RP}$

Prove: $\overline{WX} \cong \overline{ZP}$



Reasons			
Given	Combine like terms	Definition of congruence	Definition of midpoint
Division property of equality		Segment addition postulate	Substitution property of equality

	Statement	Reason
1.	W is the midpoint of $\overline{VX}$, Z is the midpoint of $\overline{RP}$, and $\overline{VX} \cong \overline{RP}$	Given
2.	$\overline{VW} \cong \overline{WX}$ $\overline{RZ} \cong \overline{ZP}$	Definition of midpoint
3.	$VX = RP$	Definition of congruence
4.	$VW + WX = VX$ $RZ + ZP = RP$	Segment addition postulate
5.	$VW = WX$ $RZ = ZP$	Definition of congruence
6.	$VW + WX = RZ + ZP$	Substitution property of equality
7.	$WX + WX = ZP + ZP$	Substitution property of equality
8.	$2WX = 2ZP$	Combine like terms
9.	$WX = ZP$	Division property of equality
10.	$\overline{WX} \cong \overline{ZP}$	Definition of congruence